Homework 1 Report

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Part 1: MOSFET Analysis

Q1.1: Analytical Current Calculation

Calculations

The saturation current for a MOSFET is given by the equation:

$$I_D = \frac{1}{2}\mu C_{ox}\frac{W}{L}(V_{GS} - V_{th})^2$$

NMOS Current Calculation: Given: $\mu_n C_{ox} = 200\mu A/V^2$, $W/L = 10$, $V_{GS} = 1.8V$, $V_{th,n} = 0.7V$.

$$I_{D,n} = 0.5 \times 200 \times 10 \times (1.8 - 0.7)^2$$

$$I_{D,n} = 1000 \times (1.1)^2 = 1000 \times 1.21$$

$$\mathbf{I_{D,n} = 1210\,\mu A}$$

PMOS Current Calculation: Given: $\mu_p C_{ox} = 80\mu A/V^2$, $W/L = 20$, $V_{SG} = 1.8V$, $|V_{th,p}| = 0.9V$.

$$I_{D,p} = 0.5 \times 80 \times 20 \times (1.8 - 0.9)^2$$

$$I_{D,p} = 800 \times (0.9)^2 = 800 \times 0.81$$

$$\mathbf{I_{D,p} = 648\,\mu A}$$

Analysis

PMOS devices typically have wider channels in digital cells because the mobility of holes ($\mu_p \approx 80$) is significantly lower than that of electrons ($\mu_n \approx 200$). To achieve symmetrical rise and fall times (equal drive current) in a CMOS gate, the PMOS width must be increased to compensate for its lower mobility.

Q1.2: NMOS $I_D - V_G$ Simulation

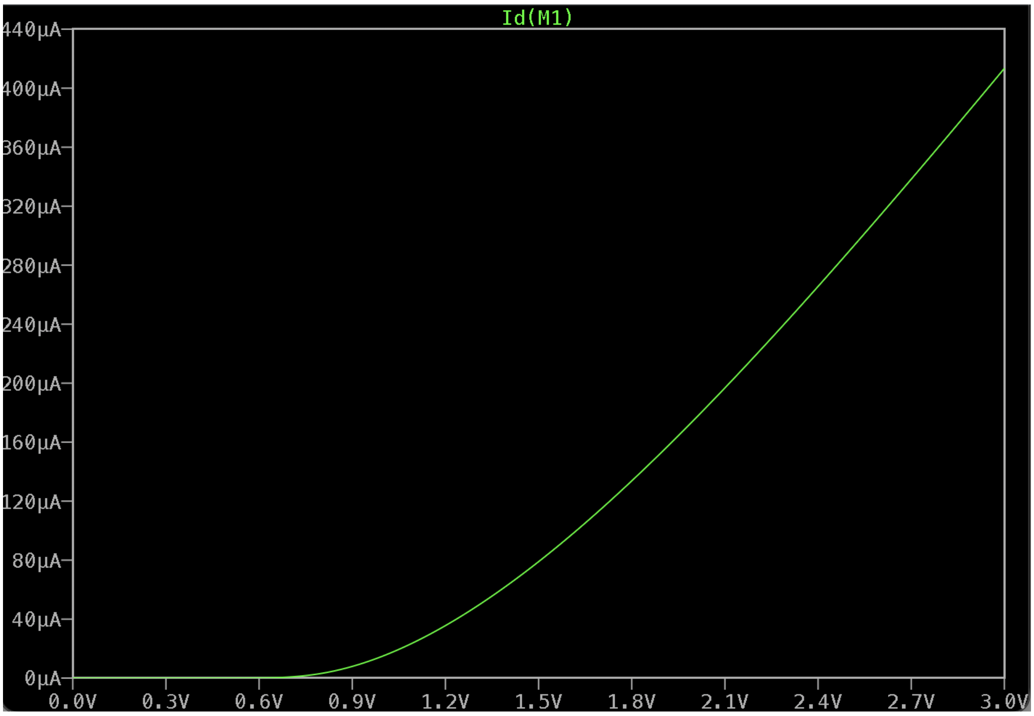

Figure 1: NMOS I_D vs V_{GS} Curve

Q1.3: PMOS $I_D - V_G$ Simulation

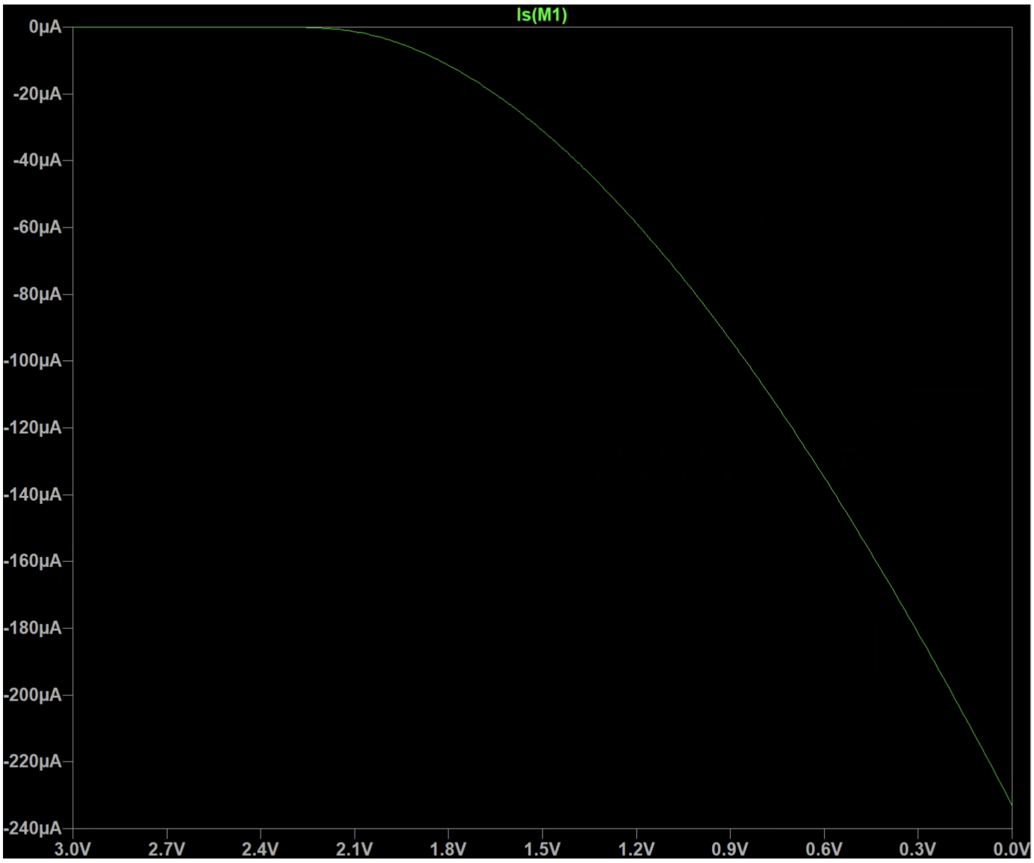

Figure 2: PMOS I_D vs V_{GS} Curve

Q1.4: Threshold Voltage Estimation

Method of Extrapolation

To estimate the threshold voltage (V_{th}), the square root of the drain current ($\sqrt{I_D}$) was plotted against the gate voltage (V_{GS}). In the saturation region, the relationship $\sqrt{I_D} \propto (V_{GS} - V_{th})$ is linear. By fitting a tangent line to the linear portion of the curve (after the inflection point) and extending it to intersect the x-axis (V_{GS} axis), the x-intercept provides the estimated V_{th}.

Results

- **Estimated NMOS V_{th}:** 0.726 V
- **Estimated PMOS V_{th}:** [Insert PMOS Value from Plot] V

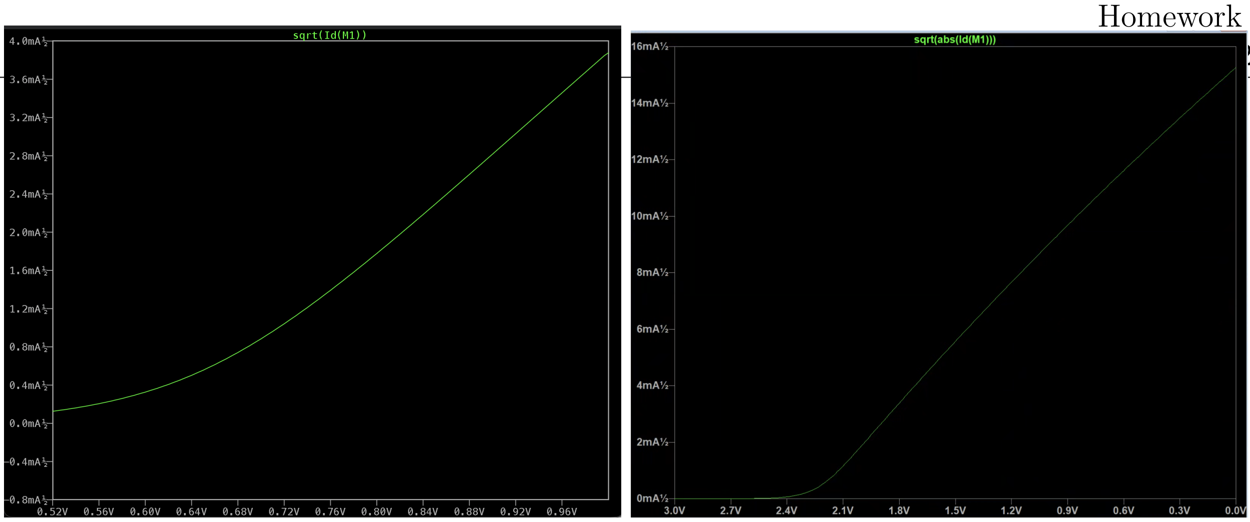

Figure 3: $\sqrt{I_D}$ vs V_{GS} plots for NMOS (left) and PMOS (right).

Q1.5: NMOS $I_D - V_D$ Simulation

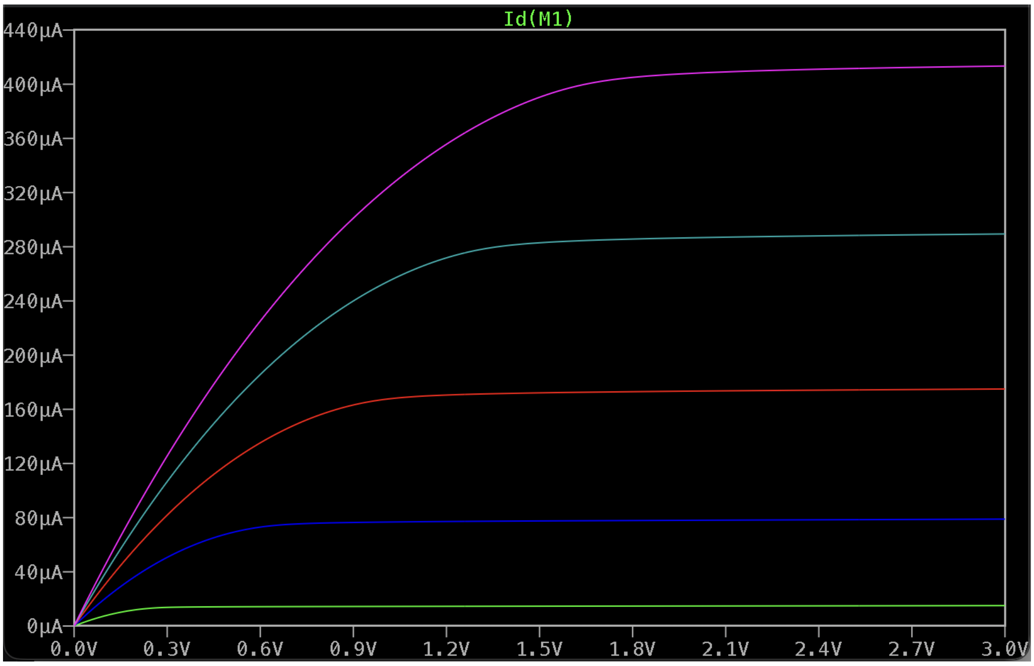

Figure 4: NMOS I_D vs V_{DS} Family of Curves

Transition Point Identification: The transition from the linear (triode) region to the saturation region occurs theoretically when $V_{DS} = V_{GS} - V_{th}$. On the waveforms, this is identified as the "knee" of the curve where the current stops increasing linearly with V_{DS} and begins to flatten out.

Part B: Inverter Analysis

Q1.6: Inverter DC Characteristic

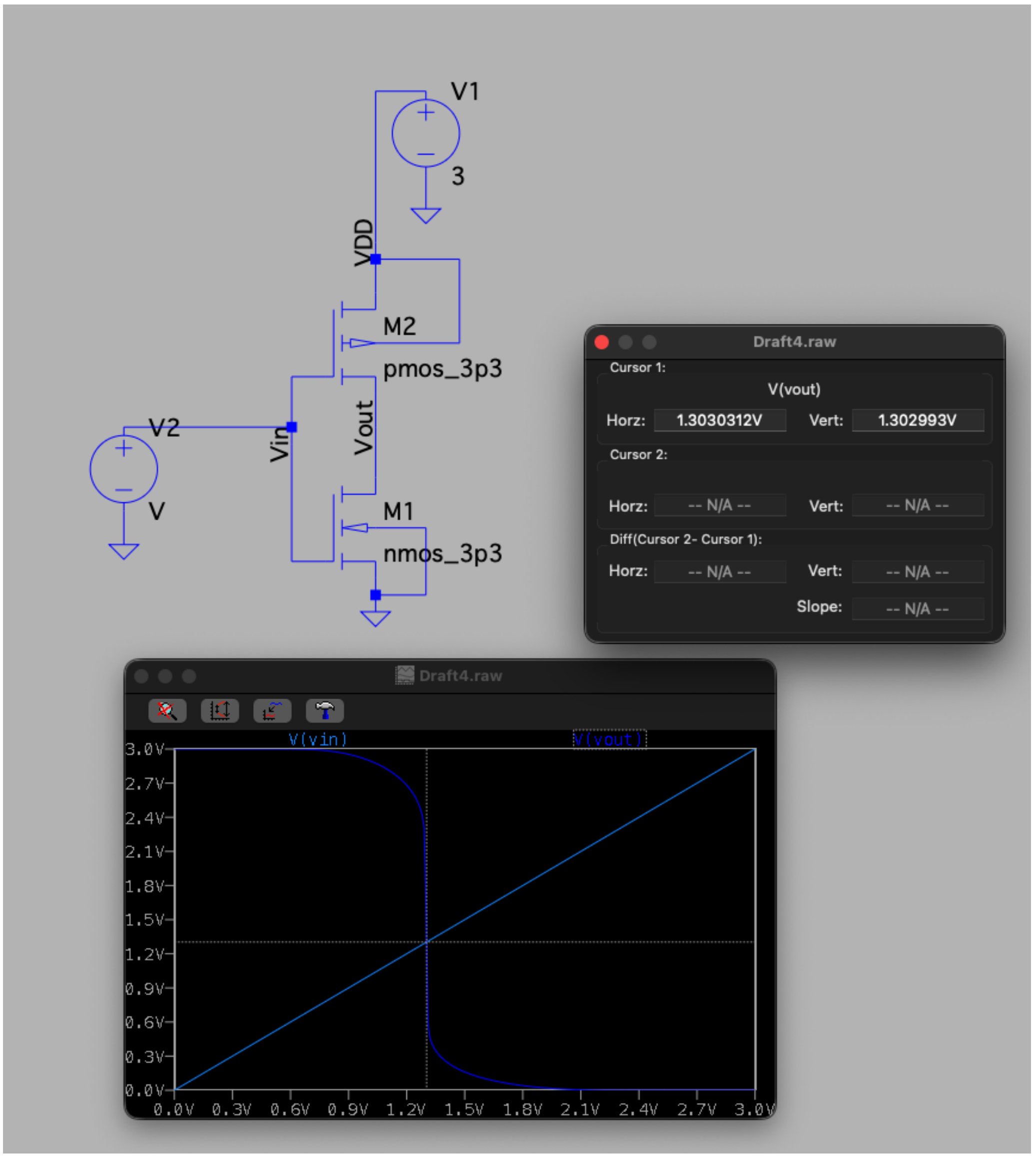

Figure 5: Inverter Voltage Transfer Characteristic (VTC)

Intersection Analysis

The intersection of V_{in} and V_{out} is located below $V_{DD}/2$ (1.5V). **Explanation:** Even though the PMOS width ($4\mu m$) is twice the NMOS width ($2\mu m$), the NMOS is still effectively "stronger." As calculated in Q1.1, the mobility of electrons ($\mu_n \approx 200$) is roughly 2.5 times higher than the mobility of holes ($\mu_p \approx 80$). Since the width ratio ($W_p/W_n = 2$) is less than the mobility ratio ($\mu_n/\mu_p \approx 2.5$), the NMOS pulls the output low faster than the PMOS pulls it high, shifting the switching threshold to a lower voltage.

Q1.7: Sizing for Symmetric Switching

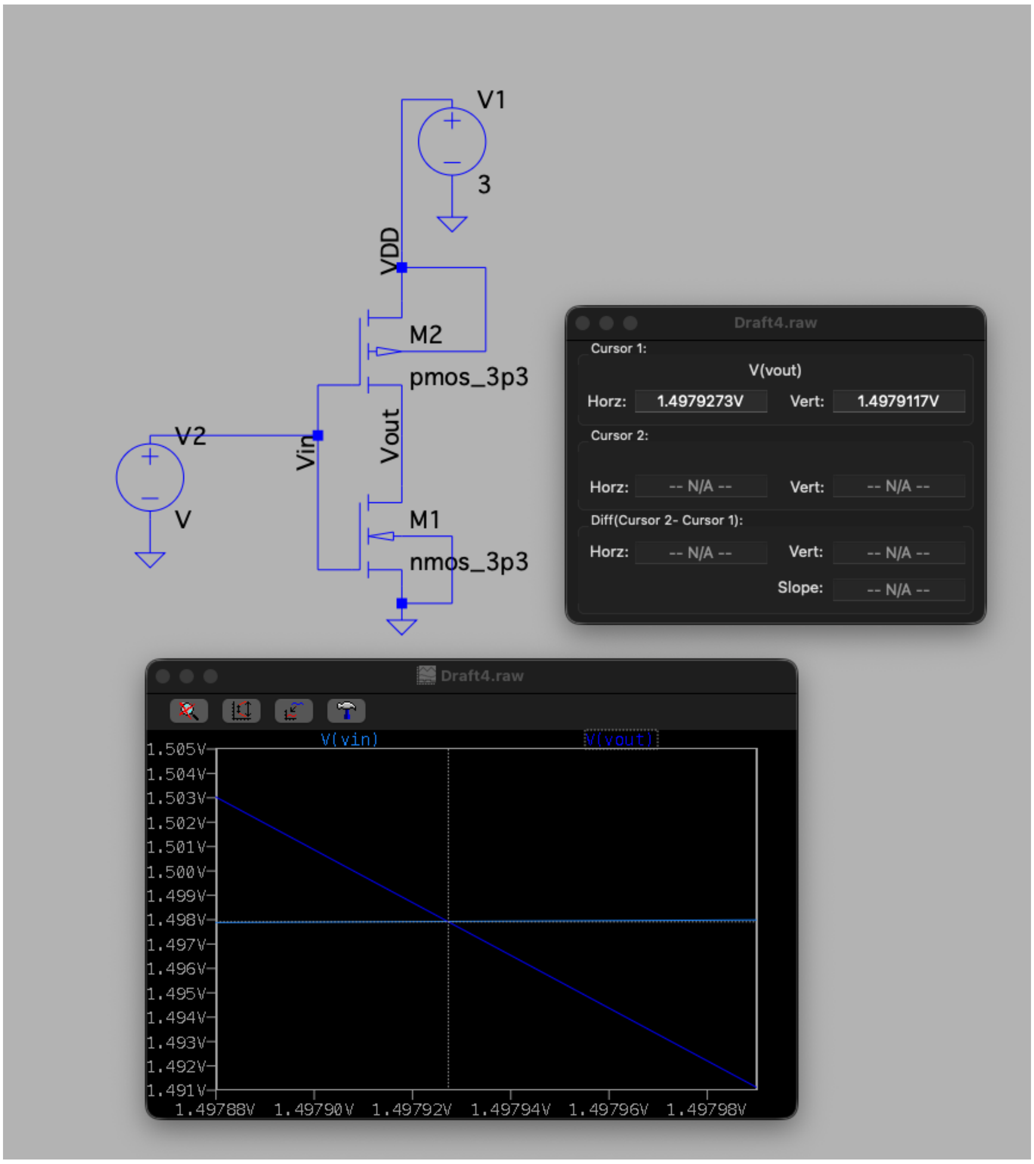

Figure 6: Inverter VTC with tuned PMOS width

To center the intersection at 1.5V ($V_{DD}/2$), the PMOS width was increased to balance the drive strengths. **Updated PMOS Sizing:** $W_p \approx 5\mu m$ (Theoretical estimate: $2\mu m \times \frac{200}{80} = 5\mu m$).

Q1.8: Transient Simulation

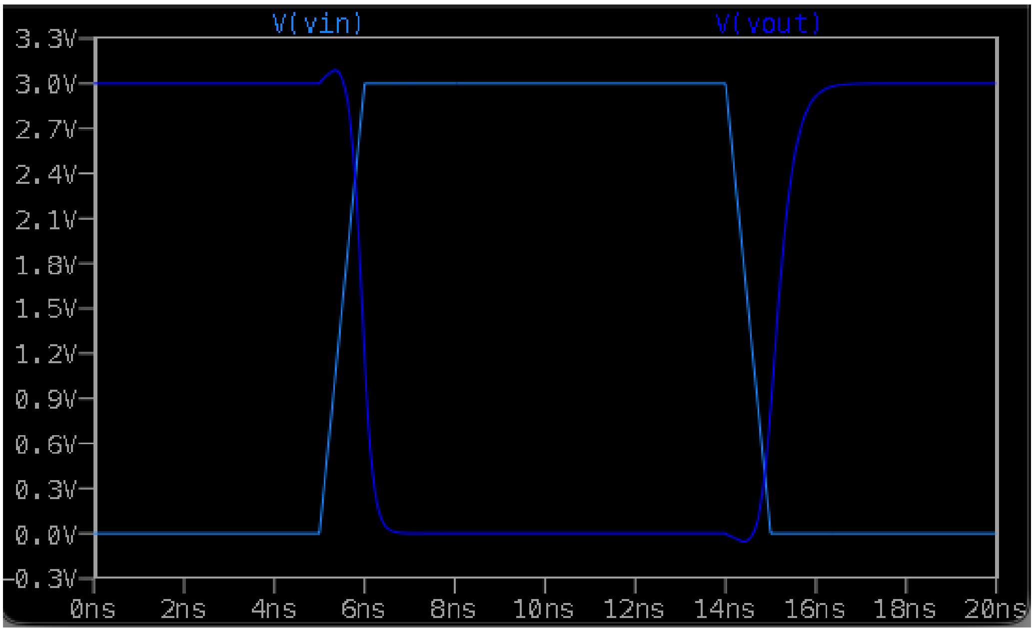

Figure 7: Transient Response (Input vs Output)

Propagation Delay Results:

- **High-to-Low Delay (t_{pHL}):** $5.9ns - 5.5ns =$ **0.4ns**
- **Low-to-High Delay (t_{pLH}):** $15.16ns - 14.49ns =$ **0.67ns**

Q1.9: Input Capacitance Estimation (Small Inverter)

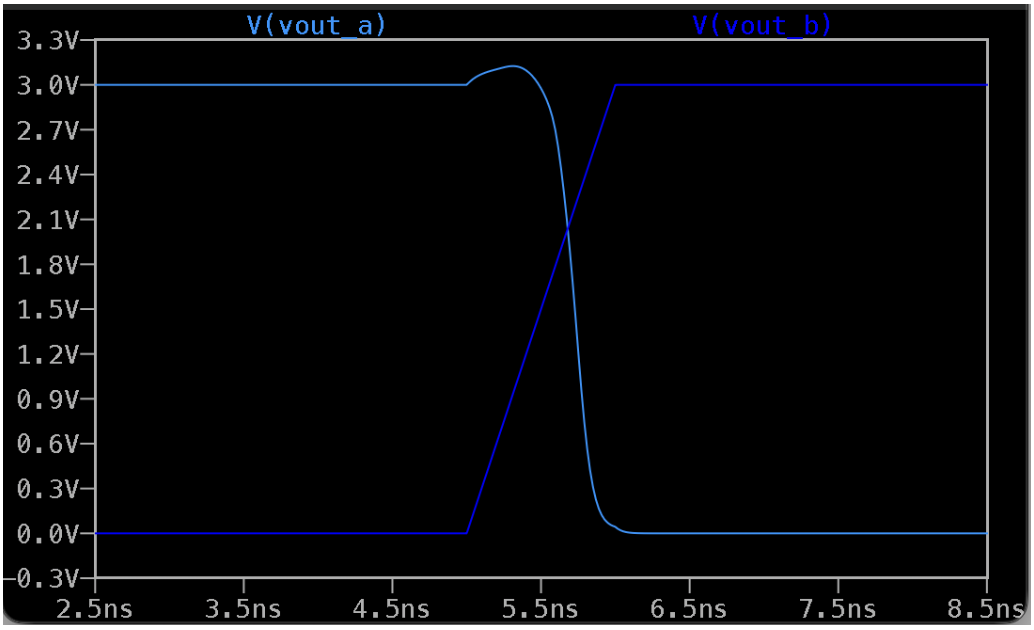

Figure 8: Delay matching for Input Capacitance Estimation

The input capacitance of Inverter C (NMOS $2\mu m$, PMOS $4\mu m$) was estimated by matching the delay of a loaded reference inverter. **Estimated C_{in}: 10 fF**

Q1.10: Input Capacitance Estimation (Large Inverter)

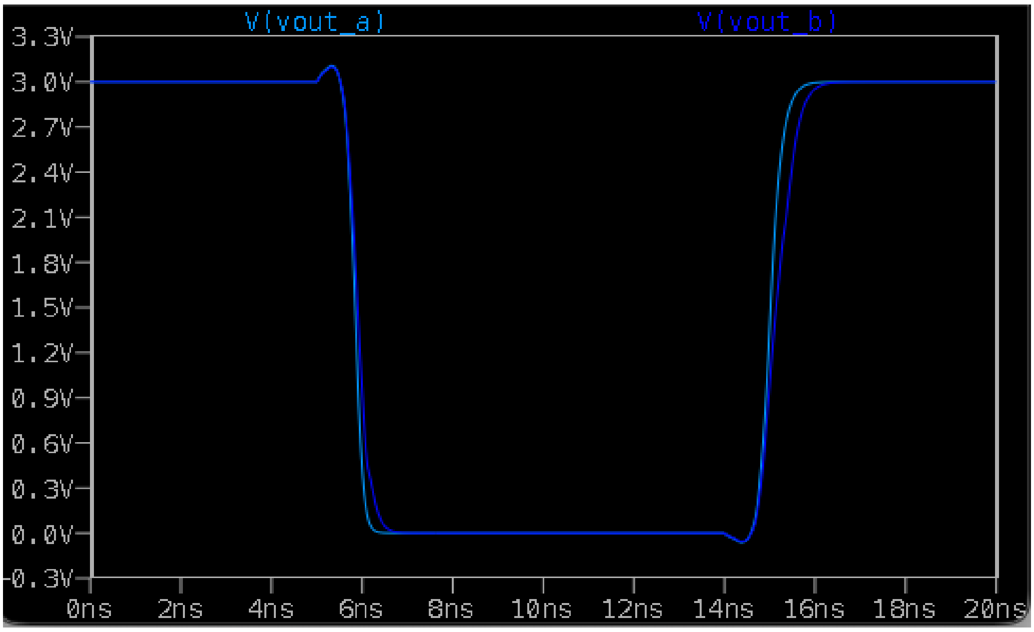

Figure 9: Delay matching for Large Inverter

The input capacitance for the larger Inverter C (NMOS $4\mu m$, PMOS $8\mu m$) was estimated. **Estimated C_{in}: 28 fF**

Explanation of Difference

The input capacitance of a MOSFET is dominated by the gate oxide capacitance, defined as $C_{gate} = C_{ox} \times W \times L$. In Q1.10, the dimensions of the inverter were doubled (Widths increased from $2\mu m/4\mu m$ to $4\mu m/8\mu m$) compared to Q1.9, while the length L remained constant. This doubling of the gate width results in a doubling of the gate area. Consequently, the input capacitance increases significantly (approx. 2.8×), reflecting the increased gate area and associated parasitic capacitances.